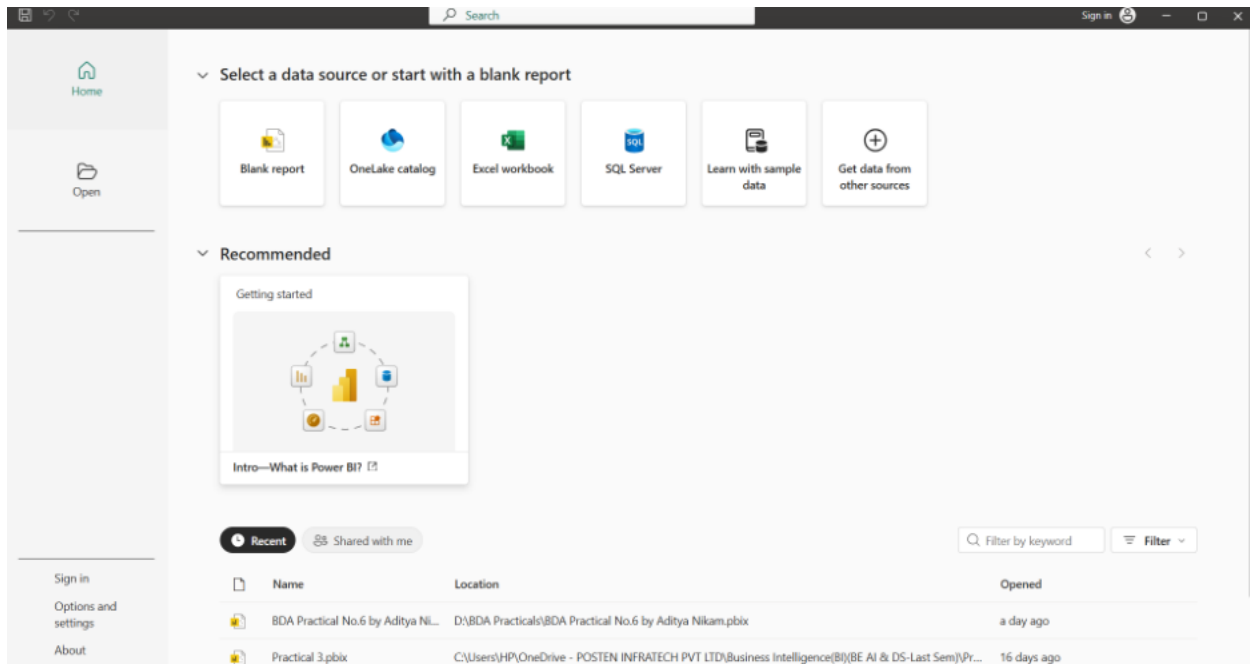


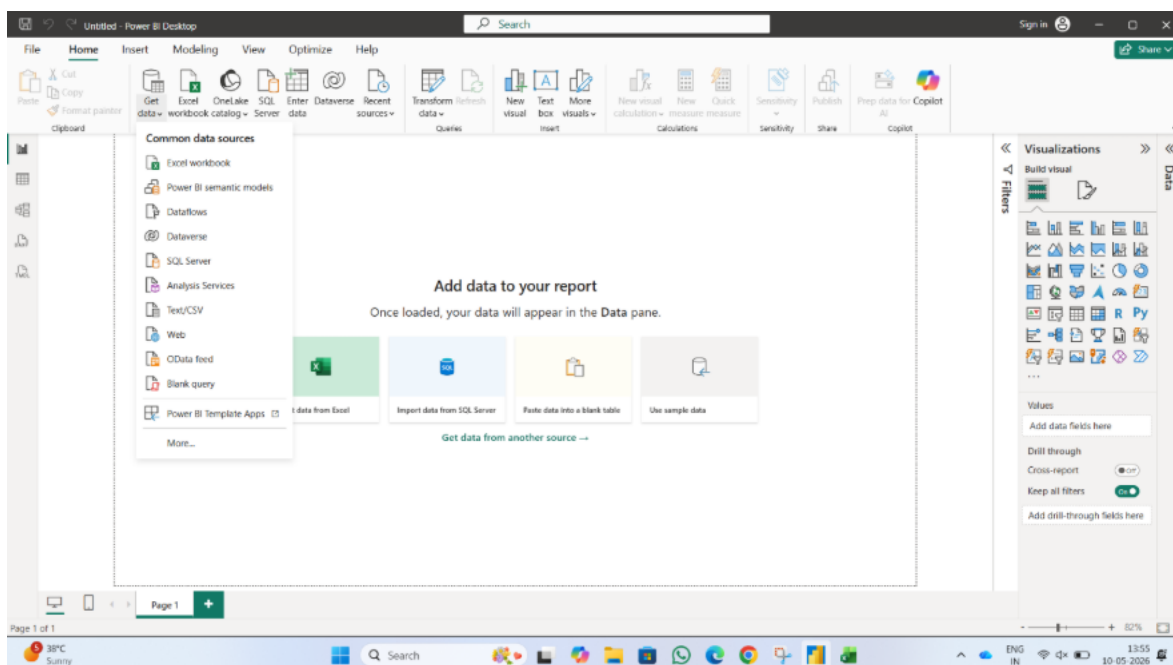
BI Practical No.2

-By Aditya Nikam(BE AI & DS)

Step 1:-Open Power BI and Click Blank Report



Step 2:-Now on the Home ribbon at the top, click on Get Data (the icon with the database and a green arrow)as your data will be .csv file so click on Excel Workbook from the drop down.



Step 3:- Now check Box.

NIFTY50_all.csv

File Origin: 1252: Western European (Windows) Delimiter: Comma Data Type Detection: Based on first 200 rows

Date	Symbol	Series	Prev Close	Open	High	Low	Last	Close	VWAP	Volume	Turnover	Trades	D
27-11-2007	MUNDRAPORT	EQ	440	770	1050	770	959	962.9	984.72	27294366	2.68772E+15		
28-11-2007	MUNDRAPORT	EQ	962.9	984	990	874	885	893.9	941.38	4581338	4.31277E+14		
29-11-2007	MUNDRAPORT	EQ	893.9	909	914.75	841	887	884.2	888.09	5124121	4.55066E+14		
30-11-2007	MUNDRAPORT	EQ	884.2	890	958	890	929	921.55	929.17	4609762	4.28326E+14		
03-12-2007	MUNDRAPORT	EQ	921.55	939.75	995	922	980	969.3	965.65	2977470	2.8752E+14		
04-12-2007	MUNDRAPORT	EQ	969.3	985	1056	976	1049	1041.45	1015.39	4849250	4.92387E+14		
05-12-2007	MUNDRAPORT	EQ	1041.45	1061	1099.5	1050	1084	1082.45	1082.79	2848209	3.08401E+14		
06-12-2007	MUNDRAPORT	EQ	1082.45	1089	1109.7	1051	1090.1	1081.3	1087.03	1749516	1.90177E+14		
07-12-2007	MUNDRAPORT	EQ	1081.3	1100	1134	1078	1100	1102.4	1106.57	2247904	2.48747E+14		
10-12-2007	MUNDRAPORT	EQ	1102.4	1110	1110	1061.1	1073.55	1075.4	1080.38	1012350	1.09373E+14		
11-12-2007	MUNDRAPORT	EQ	1075.4	1081	1089	1041	1046	1047.65	1067.8	810464	8.65416E+13		
12-12-2007	MUNDRAPORT	EQ	1047.65	1032	1065	1016	1036.9	1036.8	1043.92	744799	7.77514E+13		
13-12-2007	MUNDRAPORT	EQ	1036.8	1040	1150	1030.25	1131.15	1129.95	1109.09	3067687	3.40234E+14		
14-12-2007	MUNDRAPORT	EQ	1129.95	1139.9	1140	1101.1	1107	1110.5	1119.55	1070737	1.19875E+14		
17-12-2007	MUNDRAPORT	EQ	1110.5	1140	1168	1021.5	1052	1044.25	1102.42	1404955	1.54885E+14		
18-12-2007	MUNDRAPORT	EQ	1044.25	1045	1109.9	1031.55	1085	1074.95	1077.84	1226984	1.3225E+14		
19-12-2007	MUNDRAPORT	EQ	1074.95	1091	1116	1046.3	1078	1066.9	1082.93	845666	9.15798E+13		
20-12-2007	MUNDRAPORT	EQ	1066.9	1083.5	1083.5	1051	1067	1060.2	1065.52	623288	6.64127E+13		
24-12-2007	MUNDRAPORT	EQ	1060.2	1095	1192	1085.25	1160	1156.8	1160.77	2060892	2.39221E+14		
26-12-2007	MUNDRAPORT	EQ	1156.8	1175	1214	1148	1212	1199.9	1183.3	1467031	1.73594E+14		

Extract Table Using Examples Load Transform Data Cancel

Step 4:- Date column: Make sure it has a little **calendar icon**. If it has an "ABC" icon, click the "ABC" and change it to **Date**.

- **Symbol column:** Should be Text (ABC icon).
- **Price columns (Open, High, Low, Close, Prev Close):** Should be Decimal Numbers (1.2 icon).
- **Volume column:** Should be a Whole Number (123 icon).

Your column %Deliverble is spelled weirdly and starts with a special character, which can sometimes annoy Power BI.

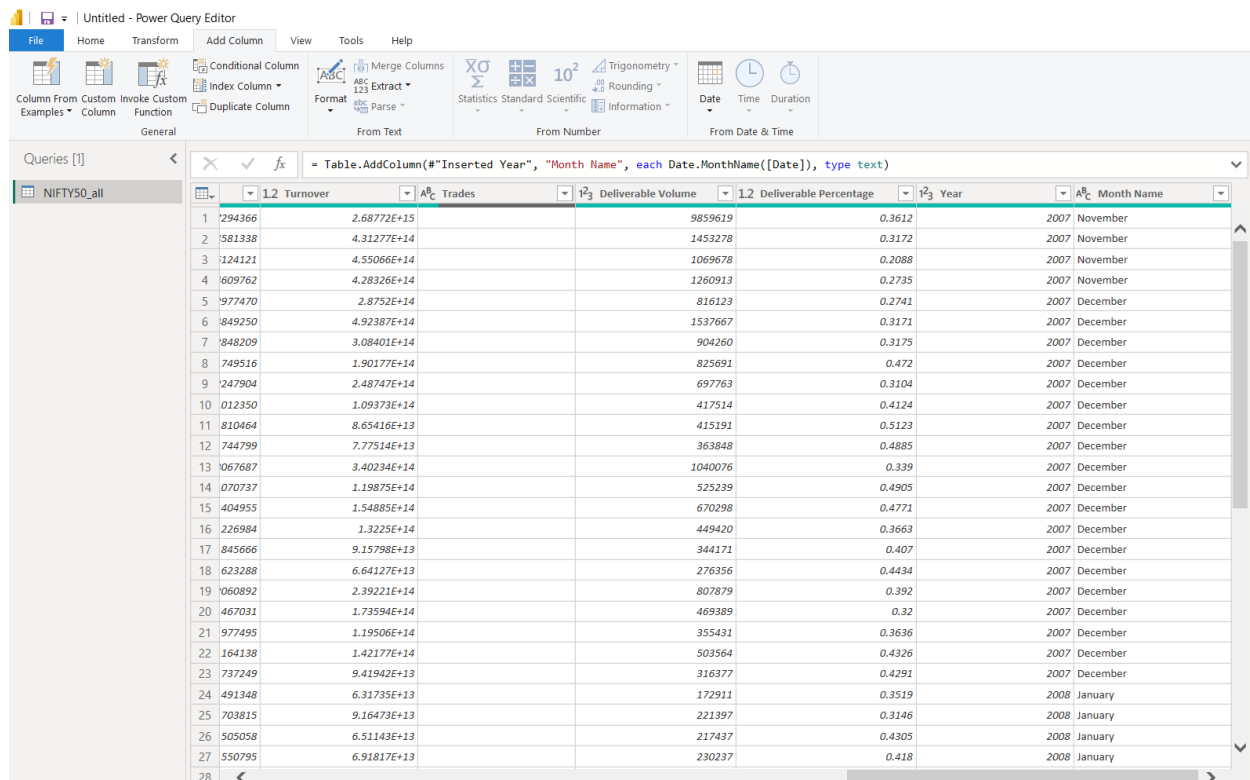
- Double-click the column header %Deliverble and type **Deliverable Percentage** to rename it.
- You can also double-click VWAP and rename it to **Avg Trade Price** so regular business users know what it means.

Step 5:-Click on the **Date** column header to highlight it. At the top of the window, click the **Add Column** tab.

Click the **Date** button (it has a little calendar and clock icon).

Go to **Year** -> **Year**. Boom! A new column appears at the far right with just the Year

Highlight the **Date** column again. Go to **Add Column** -> **Date** -> **Month** -> **Name of Month**. Now you have a column that says "November", "December", etc.



Queries [1] | Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Examples Column Invoke Custom Function Duplicate Column

Conditional Column Index Column Merge Columns Format Parse From Text

Statistics Standard Scientific Trigonometry Rounding Information

Date Time Duration From Date & Time

Table.AddColumn("#Inserted Year", "Month Name", each Date.MonthName([Date]), type text)

	1.2 Turnover	1.2 Trades	1.2 Deliverable Volume	1.2 Deliverable Percentage	1.2 Year	1.2 Month Name
1	294366	2.68772E+15	9859619	0.3612	2007	November
2	581338	4.31277E+14	1453278	0.3172	2007	November
3	124121	4.55066E+14	1069678	0.2088	2007	November
4	609762	4.28326E+14	1260913	0.2735	2007	November
5	977470	2.8752E+14	816123	0.2741	2007	December
6	849250	4.92387E+14	1537667	0.3171	2007	December
7	848209	3.08401E+14	904260	0.3175	2007	December
8	749516	1.90177E+14	825691	0.472	2007	December
9	247904	2.48747E+14	697763	0.3104	2007	December
10	012350	1.09373E+14	417514	0.4124	2007	December
11	810464	8.65416E+13	415191	0.5123	2007	December
12	744799	7.77514E+13	363848	0.4885	2007	December
13	067687	3.40234E+14	1040076	0.339	2007	December
14	070737	1.19875E+14	525239	0.4905	2007	December
15	404955	1.54885E+14	670298	0.4771	2007	December
16	226984	1.3225E+14	449420	0.3663	2007	December
17	845666	9.15798E+13	344171	0.407	2007	December
18	623288	6.64127E+13	276356	0.4434	2007	December
19	060892	2.39221E+14	807879	0.392	2007	December
20	467031	1.73594E+14	469389	0.32	2007	December
21	977495	1.19506E+14	355431	0.3636	2007	December
22	164138	1.42177E+14	503564	0.4326	2007	December
23	737249	9.41942E+13	316377	0.4291	2007	December
24	491348	6.31735E+13	172911	0.3519	2008	January
25	703815	9.16473E+13	221397	0.3146	2008	January
26	505058	6.51143E+13	217437	0.4305	2008	January
27	550795	6.91817E+13	230237	0.418	2008	January
28						

In the **Add Column** tab at the top, click **Custom Column**. A box will pop up. Name the new column: **Daily Change**. In the Custom Column Formula box, double-click **Close** from the list on the right, type a minus sign -, and double-click **Open**. The formula should look like this: [Close] - [Open] Click OK. Change the data type of this new column to a Decimal Number (1.2).

fx = Table.TransformColumnTypes(#"Added Custom",{{"Daily Change", type number}})							
	Trades	Deliverable Volume	Deliverable Percentage	Year	Month Name	Daily Change	
1	72E+15	9859619	0.3612	2007	November	192.9	
2	77E+14	1453278	0.3172	2007	November	-90.1	
3	66E+14	1069678	0.2088	2007	November	-24.8	
4	26E+14	1260913	0.2735	2007	November	31.55	
5	52E+14	816123	0.2741	2007	December	29.55	
6	87E+14	1537667	0.3171	2007	December	56.45	
7	01E+14	904260	0.3175	2007	December	21.45	
8	77E+14	825691	0.472	2007	December	-7.7	
9	47E+14	697763	0.3104	2007	December	2.4	
10	73E+14	417514	0.4124	2007	December	-34.6	
11	16E+13	415191	0.5123	2007	December	-33.35	
12	14E+13	363848	0.4885	2007	December	4.8	
13	34E+14	1040076	0.339	2007	December	89.95	
14	75E+14	525239	0.4905	2007	December	-29.4	
15	85E+14	670298	0.4771	2007	December	-95.75	
16	25E+14	449420	0.3663	2007	December	29.95	
17	98E+13	344171	0.407	2007	December	-24.1	
18	27E+13	276356	0.4434	2007	December	-23.3	
19	21E+14	807879	0.392	2007	December	61.8	
20	94E+14	469389	0.32	2007	December	24.9	
21	06E+14	355431	0.3636	2007	December	-3.35	
22	77E+14	503564	0.4326	2007	December	59.7	
23	42E+13	316377	0.4291	2007	December	5.45	
24	35E+13	172911	0.3519	2008	January	17.85	
25	73E+13	221397	0.3146	2008	January	-2.8	
26	43E+13	217437	0.4305	2008	January	-29.2	
27	17E+13	230237	0.418	2008	January	-38.45	
28							

mn profiling based on top 1000 rows

Custom Column

Add a column that is computed from the other columns.

New column name

Daily Change

Custom column formula ⓘ

=

Available columns

Date
Symbol
Series
Prev Close
Open
High
Low

<< Insert

[Learn about Power Query formulas](#)

⚠ The formula is incomplete.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

Daily Change

Custom column formula ⓘ

= [Close] - [Open]

Available columns

Prev Close
Open
High
Low
Last
Close
Average Trade Price

<< Insert

[Learn about Power Query formulas](#)

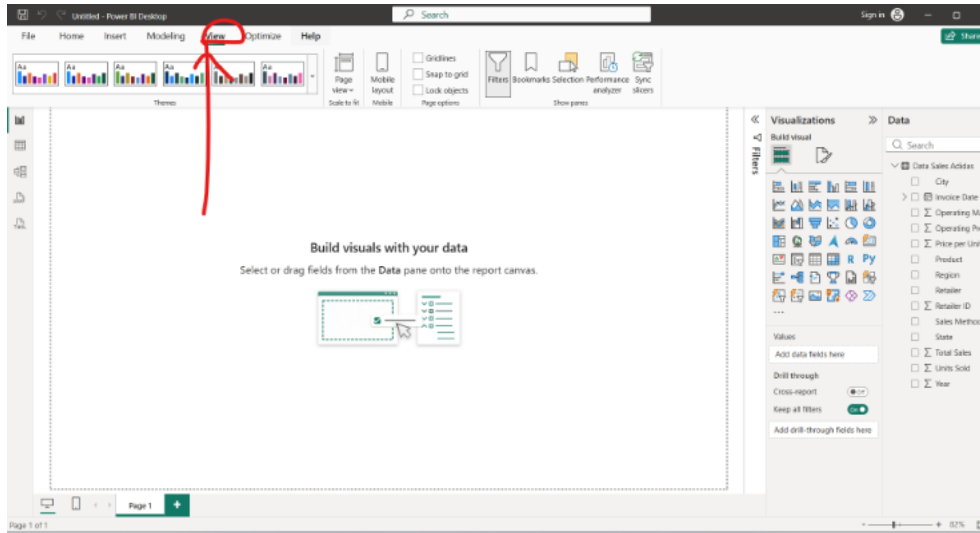
✓ No syntax errors have been detected.

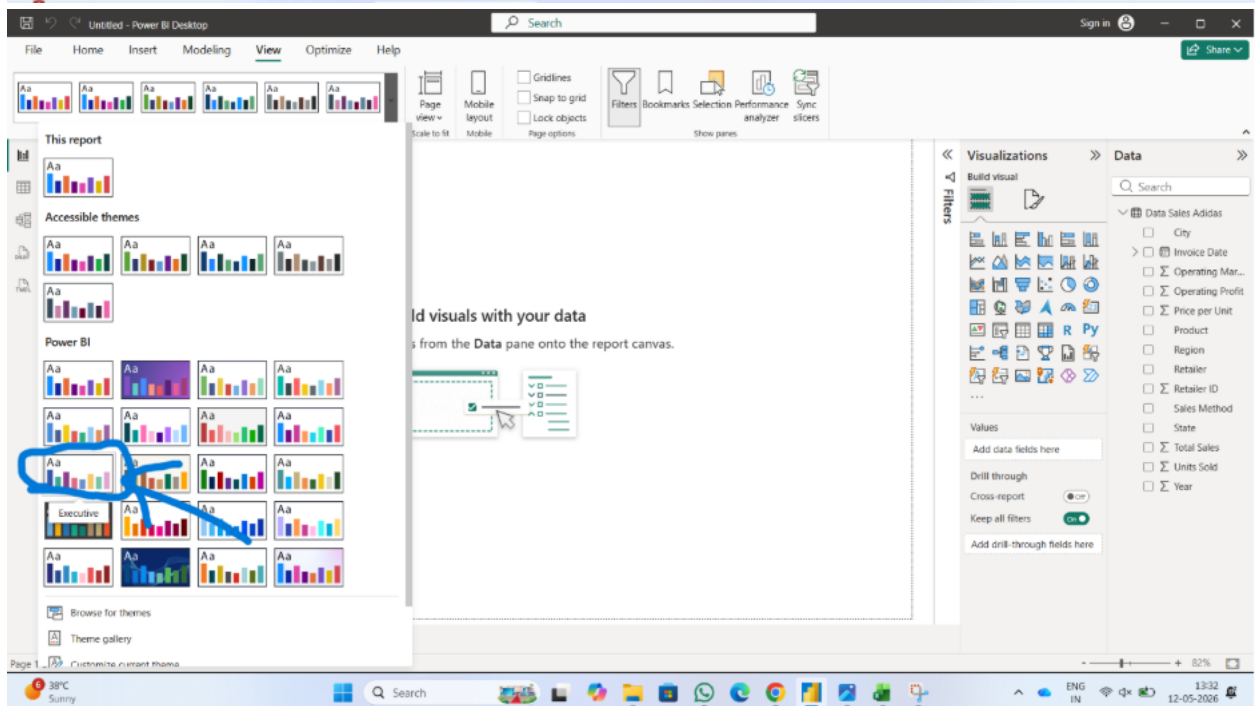
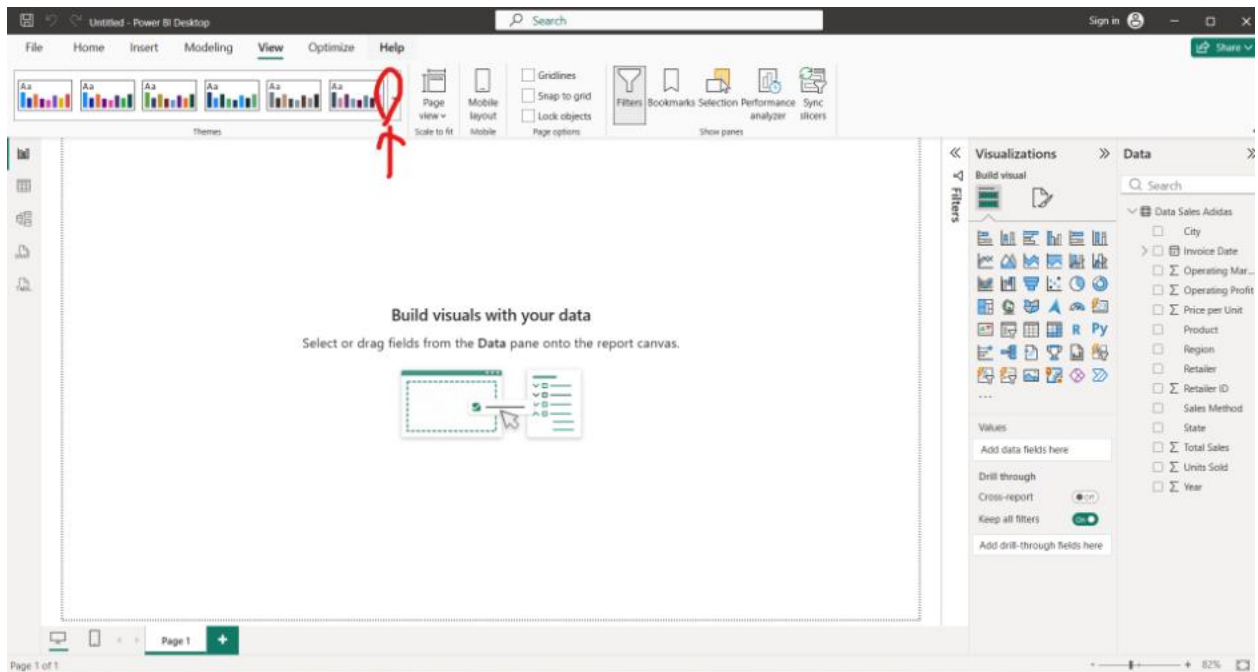
OK

Cancel

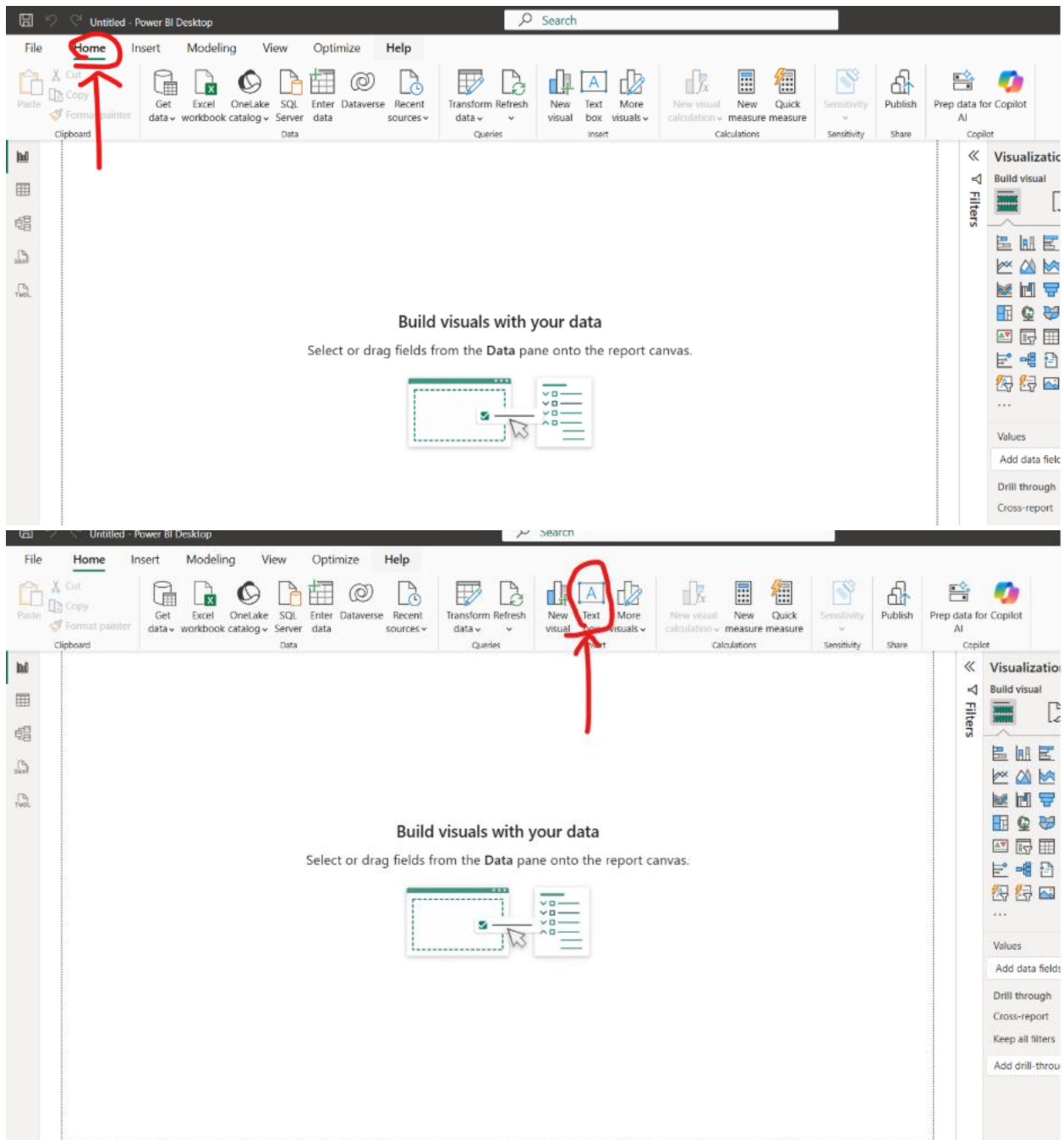
Step 6:- At the top left of the Power Query Editor, click the Home tab. Click the big Close & Apply button (the icon with a purple disk and a door).

Step No-7:- Go to the View tab at the top of the screen. You will see a gallery of "Themes" (colored boxes). Click the dropdown arrow to see them all. Choose a dark theme (like "Innovate" or "Executive") or a sleek light one. This automatically styles your charts!

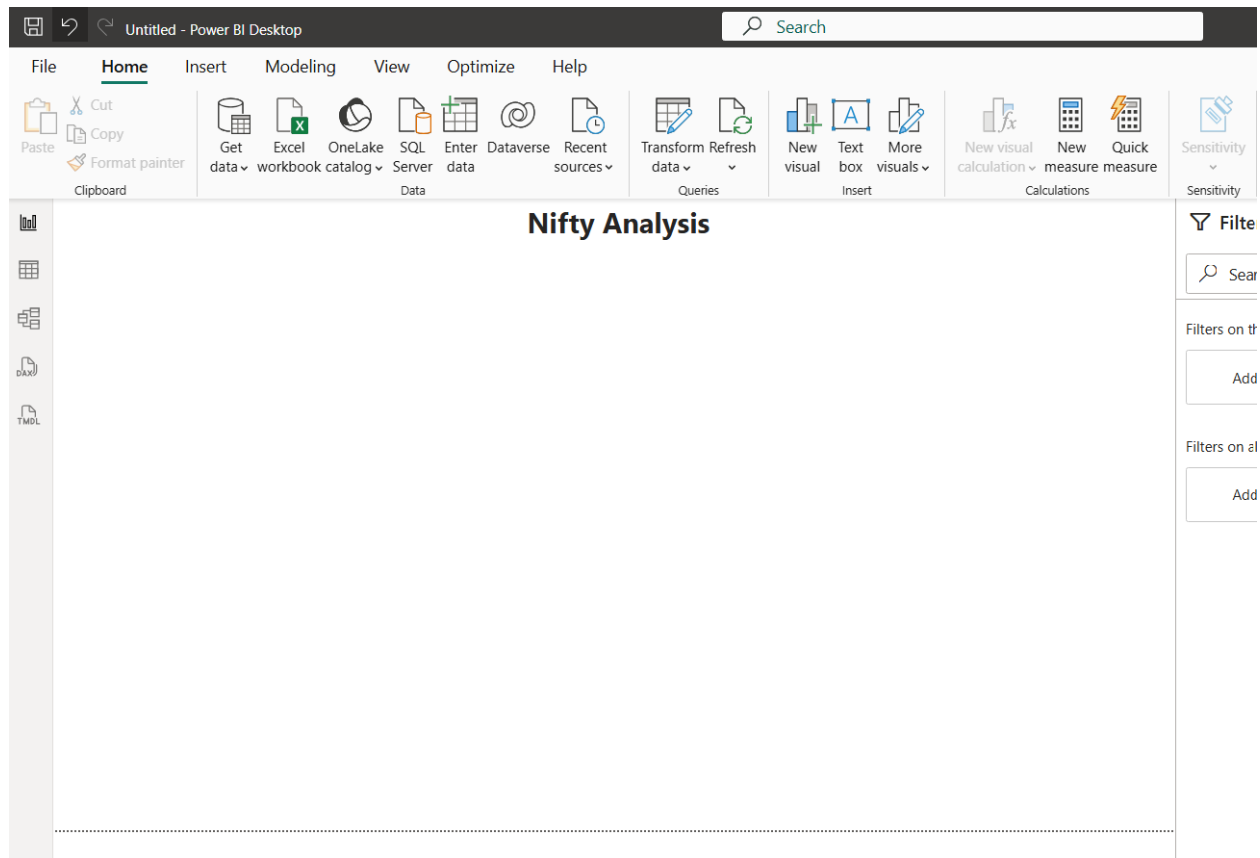




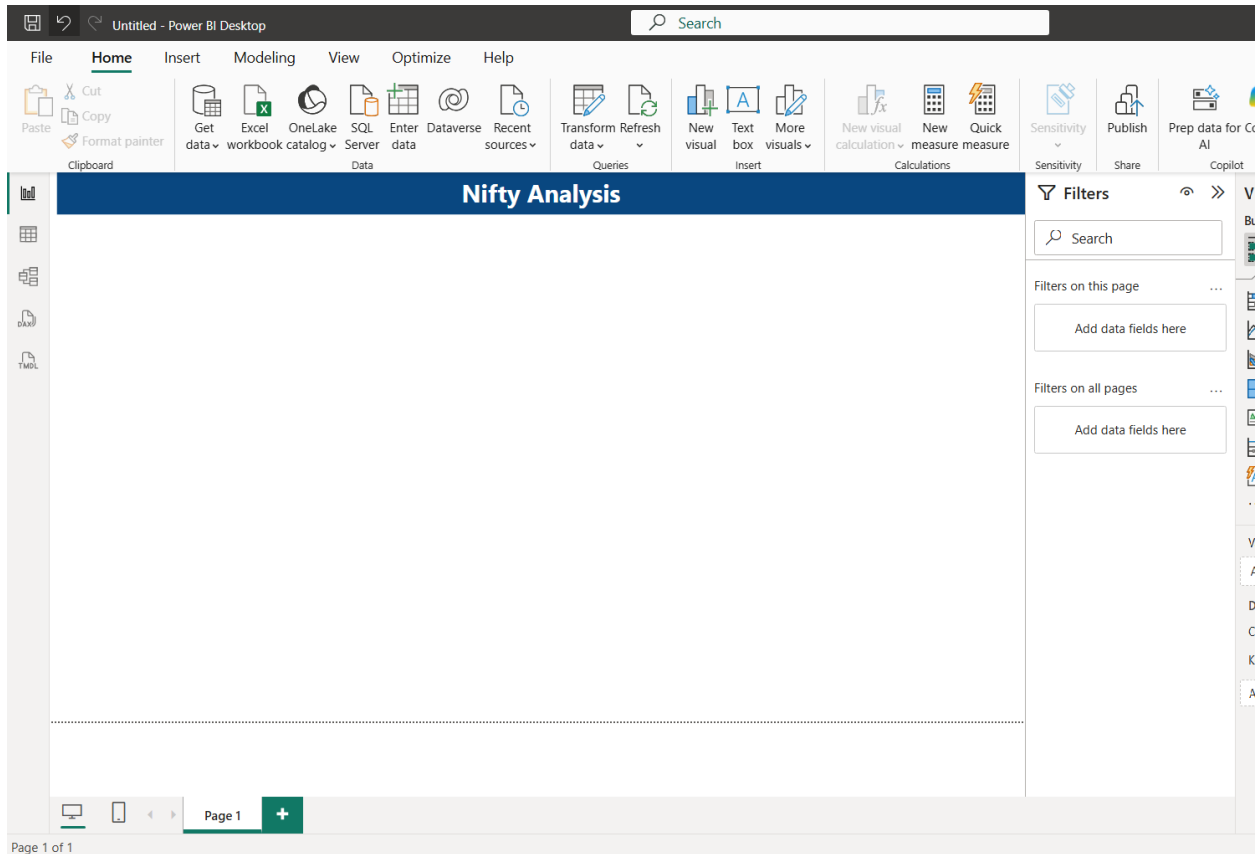
Step No-8:- Go to the Insert tab at the top Click Text box.



Step No-9:-Now type “Nifty Analysis” and increase font size to 24 and make the text Bold and make the box wider and smaller in height and keep the text centered aligned.



Step 10:-Now Click on the background of the text “Nifty Analysis” and then in Format Text Box Expand the “Effects” section and change the background color to Navy Blue



Step No:11:- Look at the **Visualizations** pane on the right side. Click the visual that looks like a little funnel with a checklist (it's called a **Slicer**). An empty box will appear on your canvas. Look at your **Data** pane on the far right. Find the **Symbol** column and drag it into the empty Slicer box.

Visualizations >> Data

Build visual

Search

Visualizations grid:

- Bar charts, Line charts, Pie charts, Donut charts, Maps, Tables, R, Py, and other visualization types.

Values

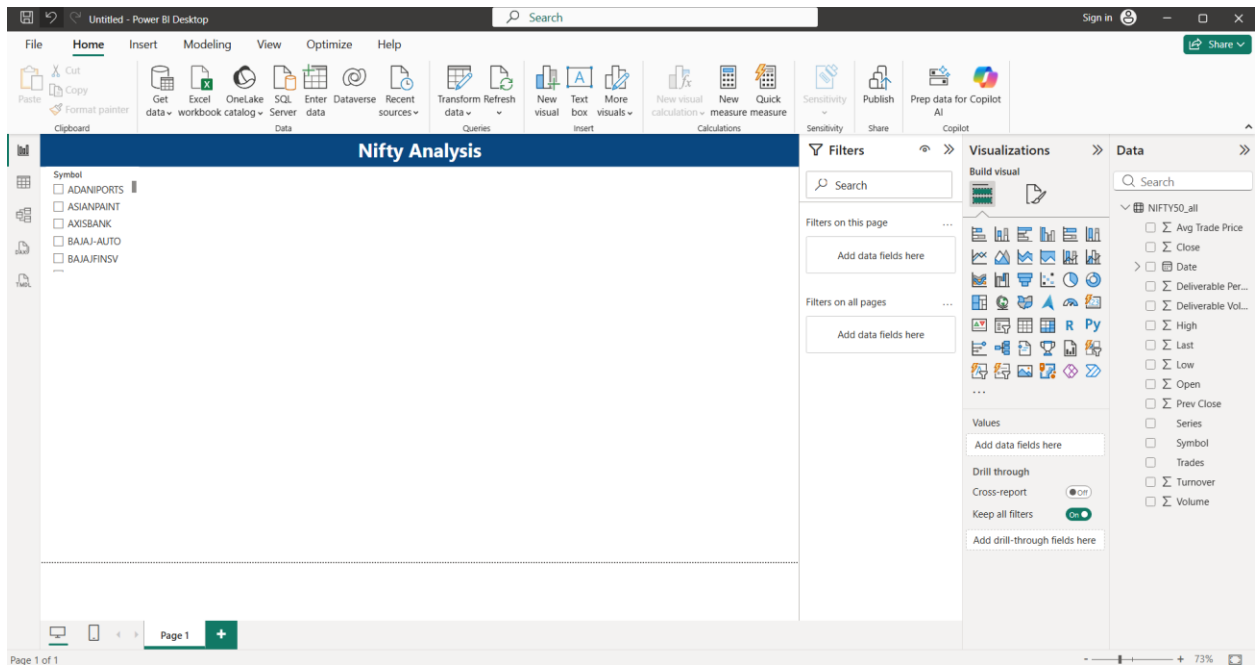
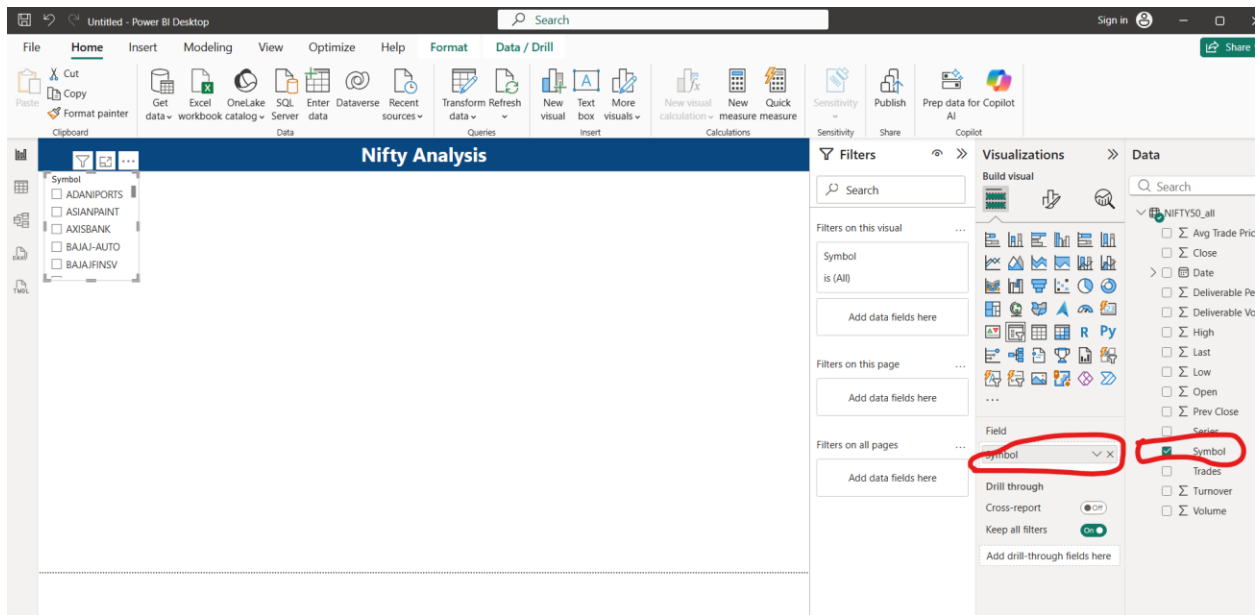
Add data fields here

Drill through

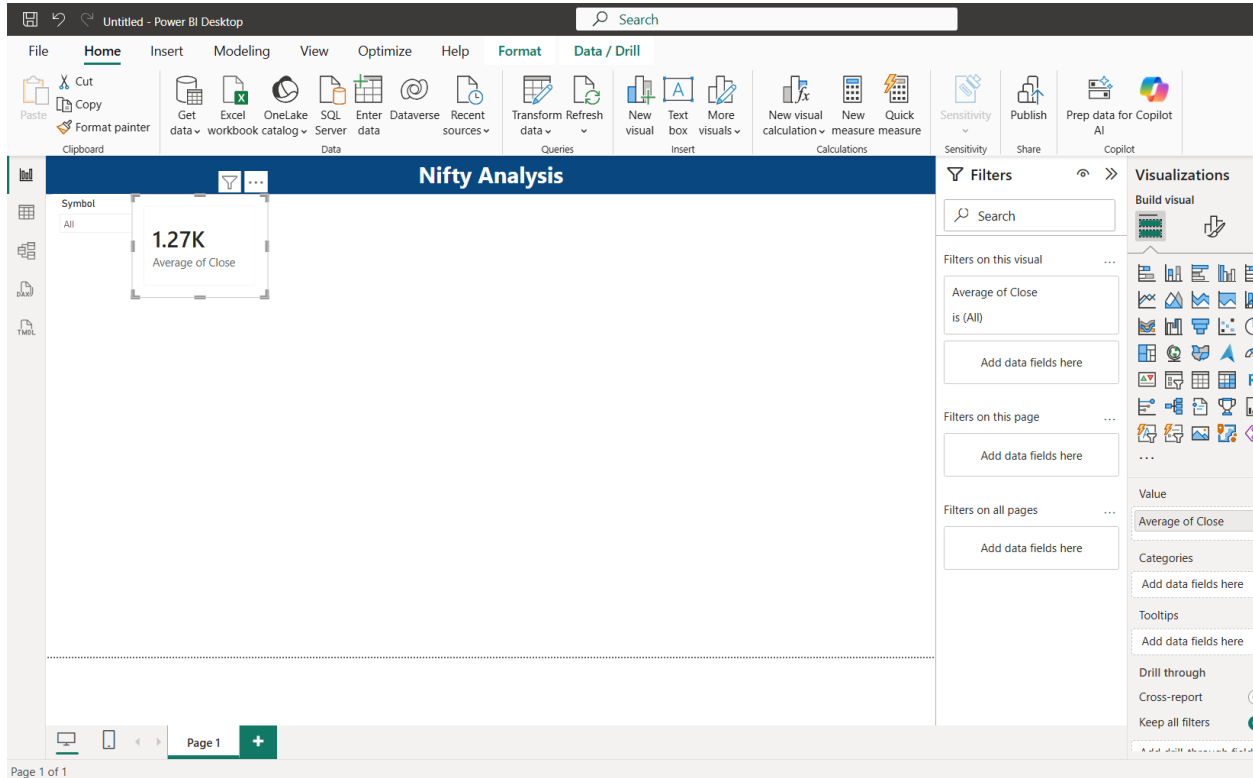
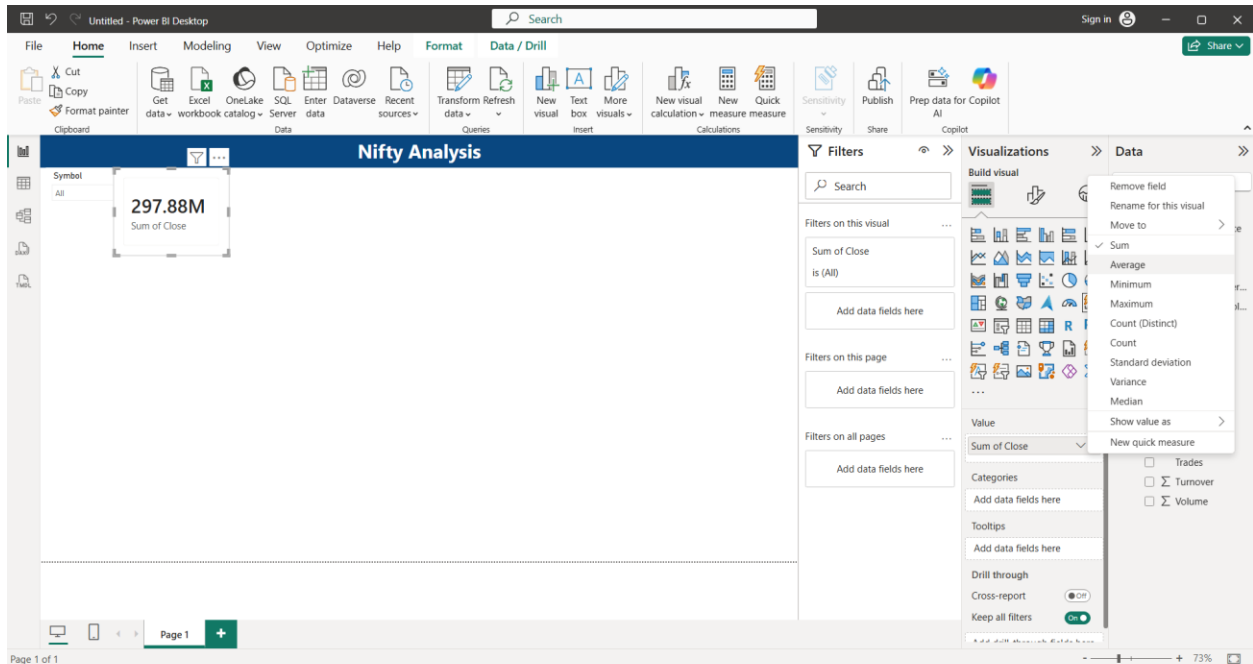
Cross-report ☐ Off

Keep all filters ☒ On

Add drill-through fields here



Step No 11:- In the Visualizations pane, click the visual that has the numbers "123" on it (Card). From the Data pane, drag the **Close** column into this card.



Step 12:- Go to the Format paintbrush under Visualization Pane.Go to General -> Title, turn it on, and name it "Average Closing Price".

ers

arch

this visual

of Close

Id data fields here

this page

Id data fields here

all pages

Id data fields here

Visualizations

Format visual

Search

Visual General

Cards

Callout

Apply settings to

Cards

All

Layout

Vertical alignment

Callout size

50 %

Padding

Value

On

Data

Search

NIFTY50_all

Avg Trade Price

Close

Date

Deliverable Per...

Deliverable Vol...

High

Last

Low

Open

Prev Close

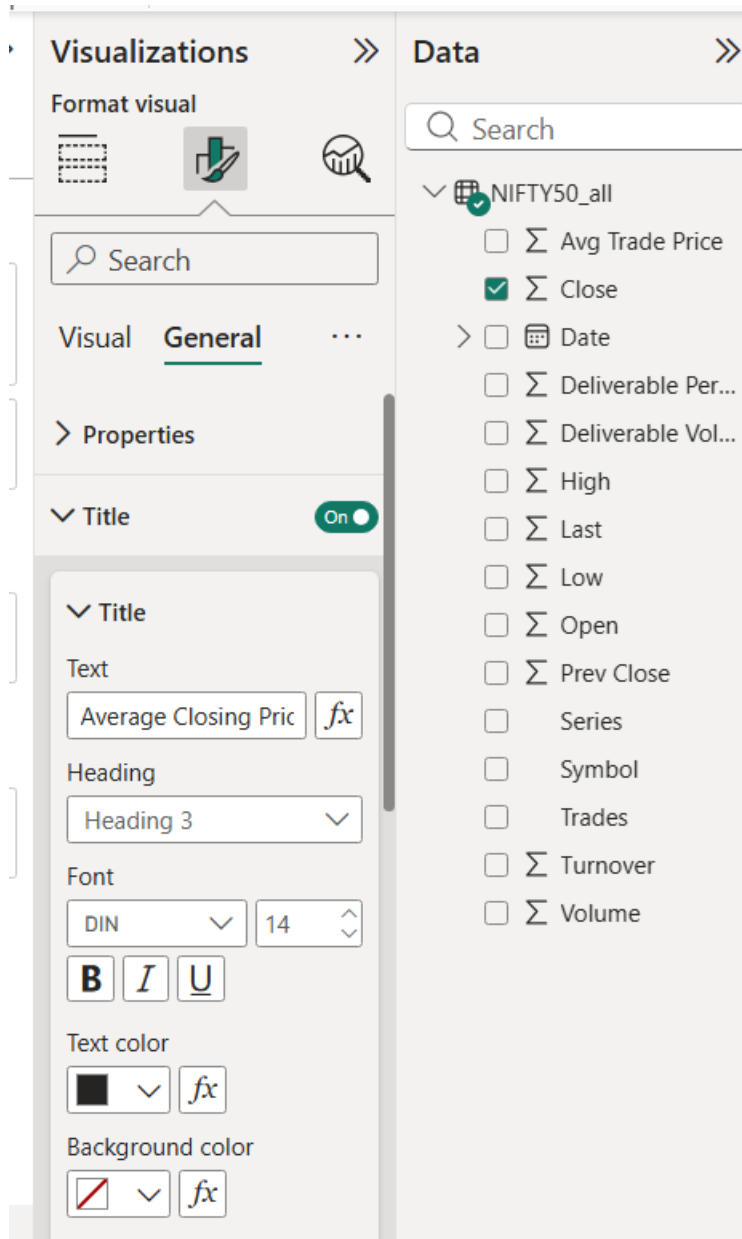
Series

Symbol

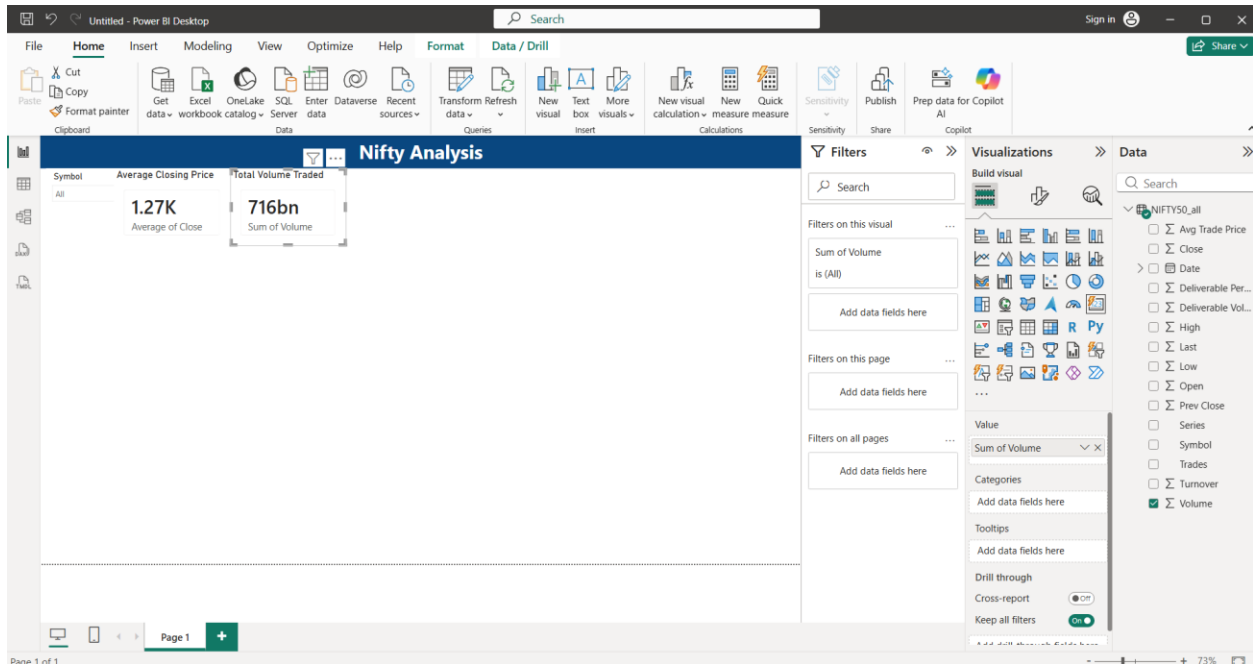
Trades

Turnover

Volume



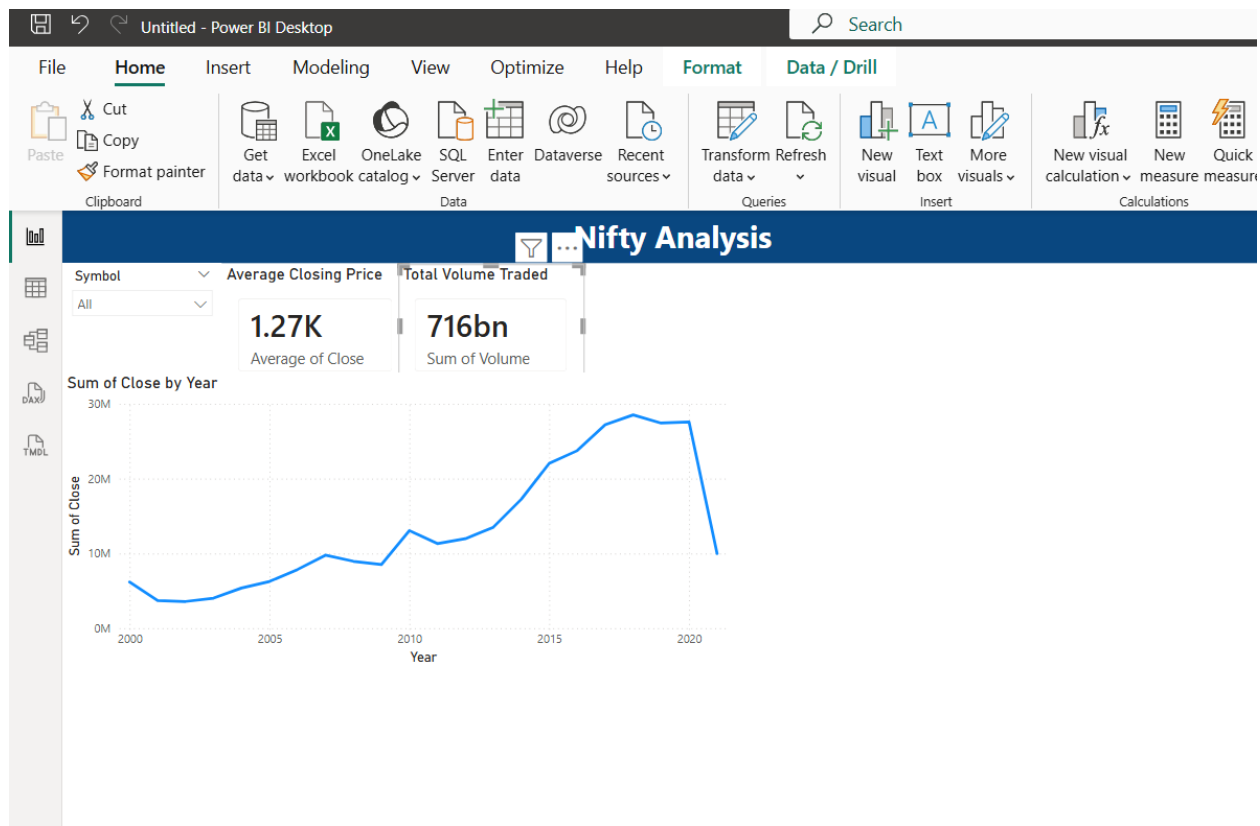
Step 13- Create one more Card: Copy and paste the card you just made. Replace "Close" with your **Volume** column in Value field. Make sure it is set to **Sum**. Change the title to "Total Volume Traded"



Step 14- Click anywhere on the blank canvas. In the Visualizations pane, click the **Line chart** icon (it looks like a jagged line going up).

Make this chart very large—it should take up the middle of your screen. Drag your **Date** column into the **X-axis** box in the Visualizations pane.

Drag your **Close** column into the **Y-axis** box. If the chart looks like a flat line with just a few dots, look at the top right corner of the visual itself.



Step 15:- Click anywhere on the **blank white space** of your dashboard canvas (make sure no other charts are currently highlighted/selected).

Look at the **Visualizations pane** on the right side of your screen.

Find the icon that looks like a ring or a donut with a hole in the middle (it will say **Donut chart** when you hover your mouse over it).

Click that icon. An empty, square placeholder for your Donut chart will appear on the canvas.

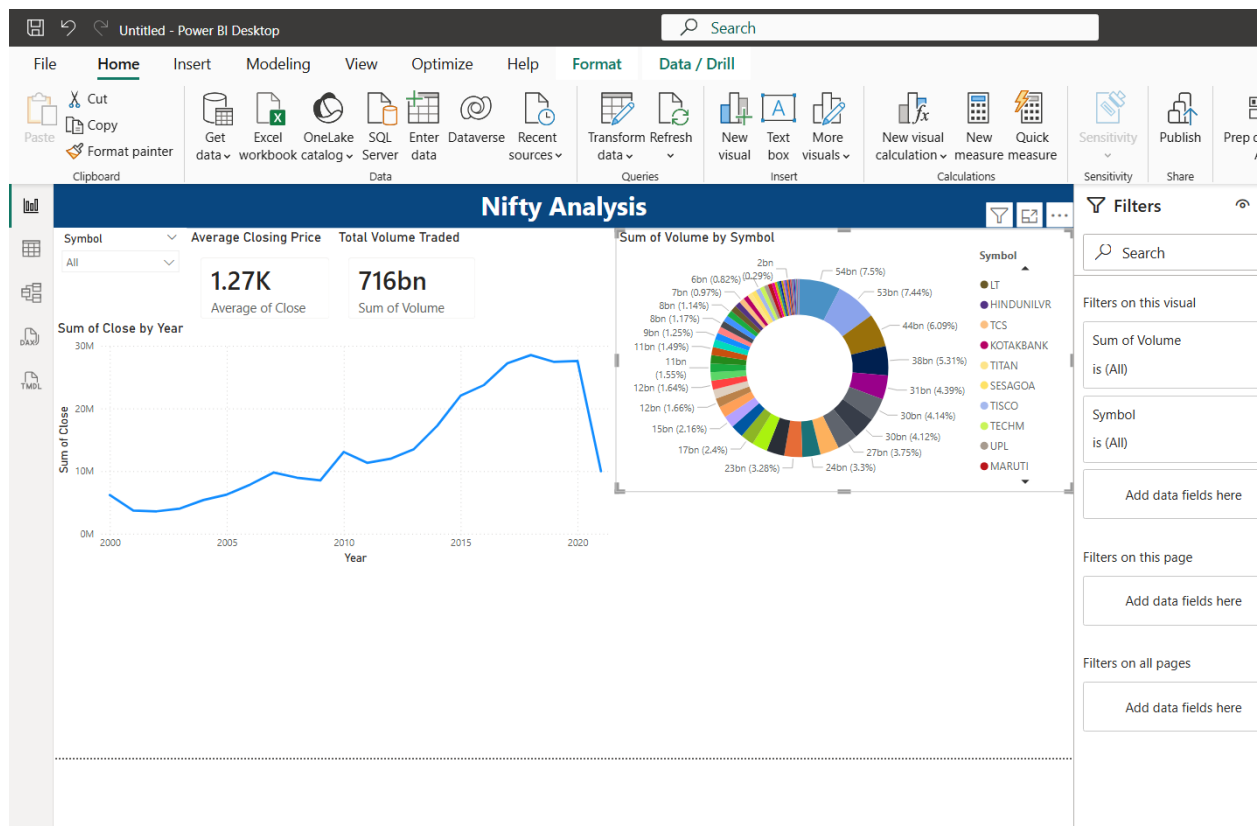
Click and drag the edges of this box to resize it, and move it to the bottom left of your dashboard.

Go to your **Data pane** on the far right (where all your columns are listed).

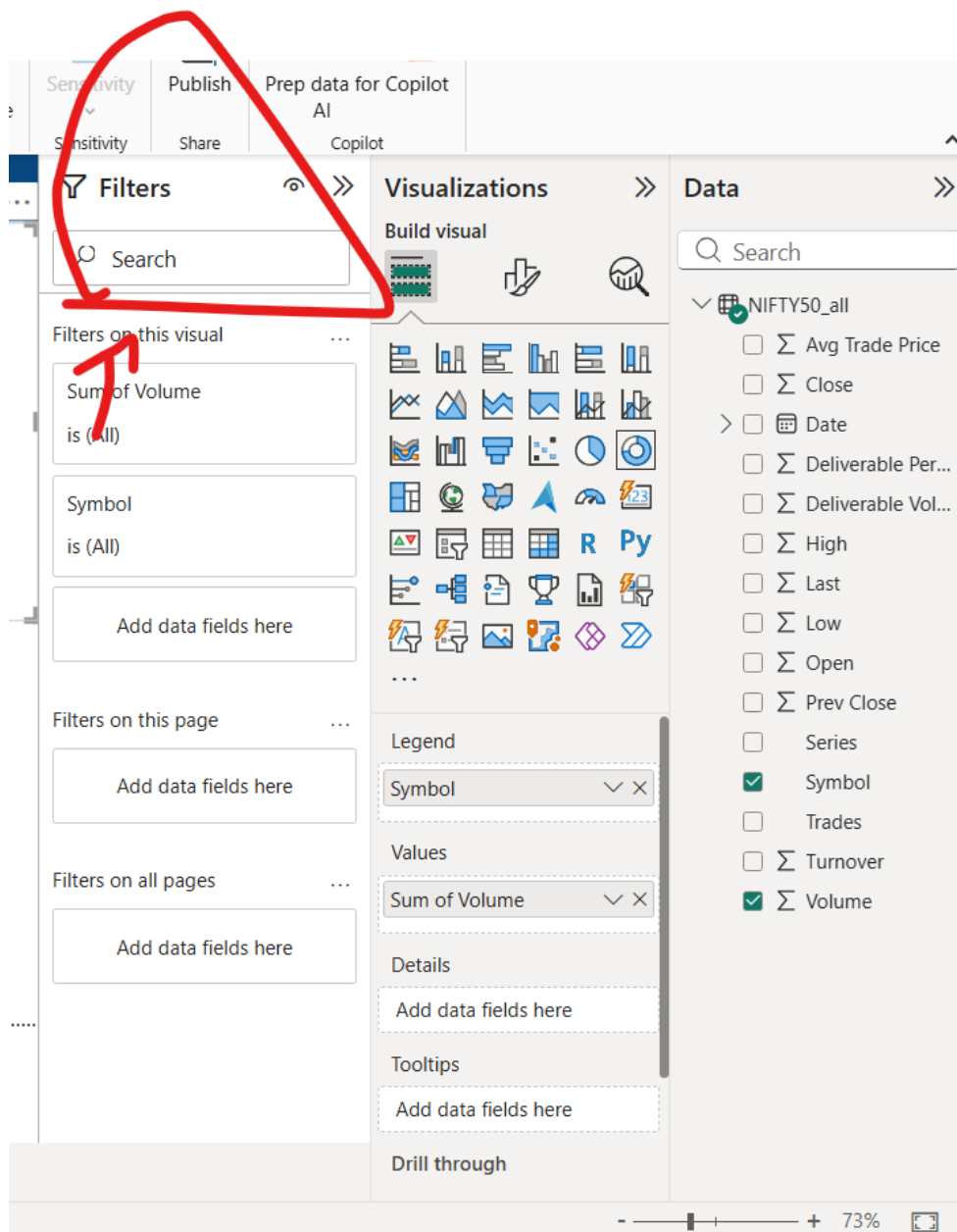
Click and hold the **Symbol** column, drag it over, and drop it exactly into the **Legend** box.

Next, click and hold the **Volume** column, drag it over, and drop it exactly into the **Values** box.

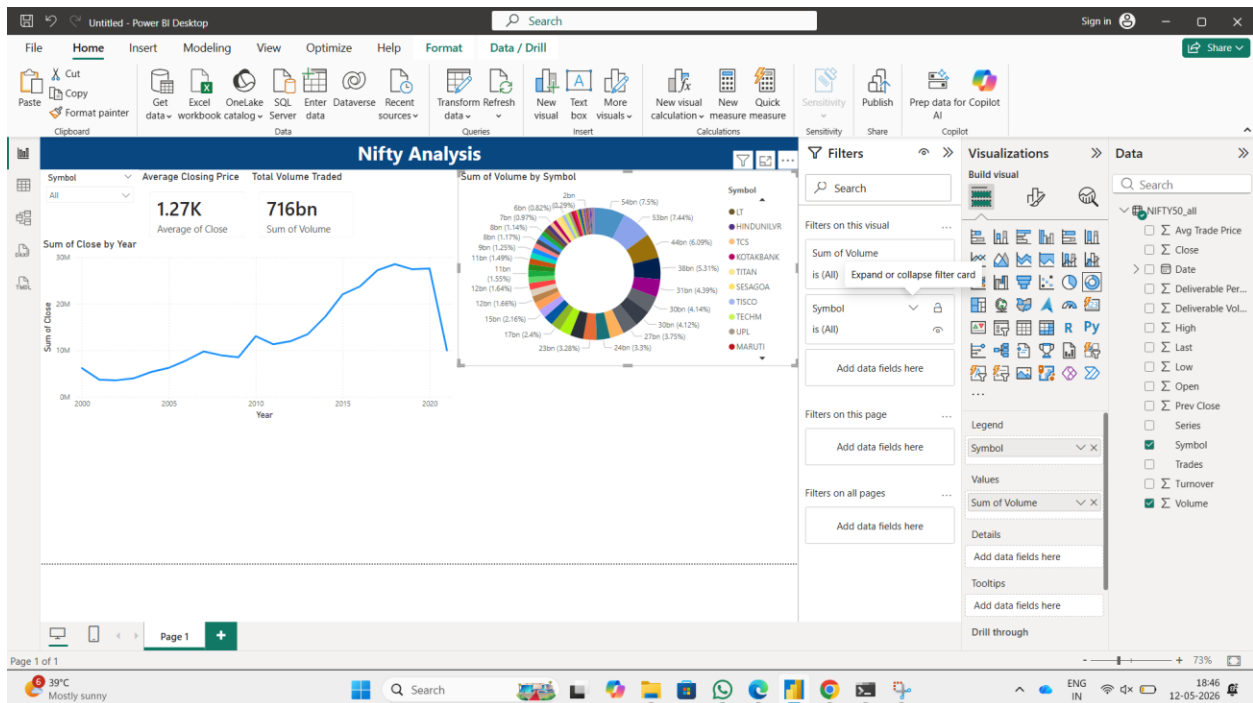
At this moment, your donut will look like a messy, colorful wheel with 50 tiny slices because it's trying to show every single stock at once. Let's clean that up



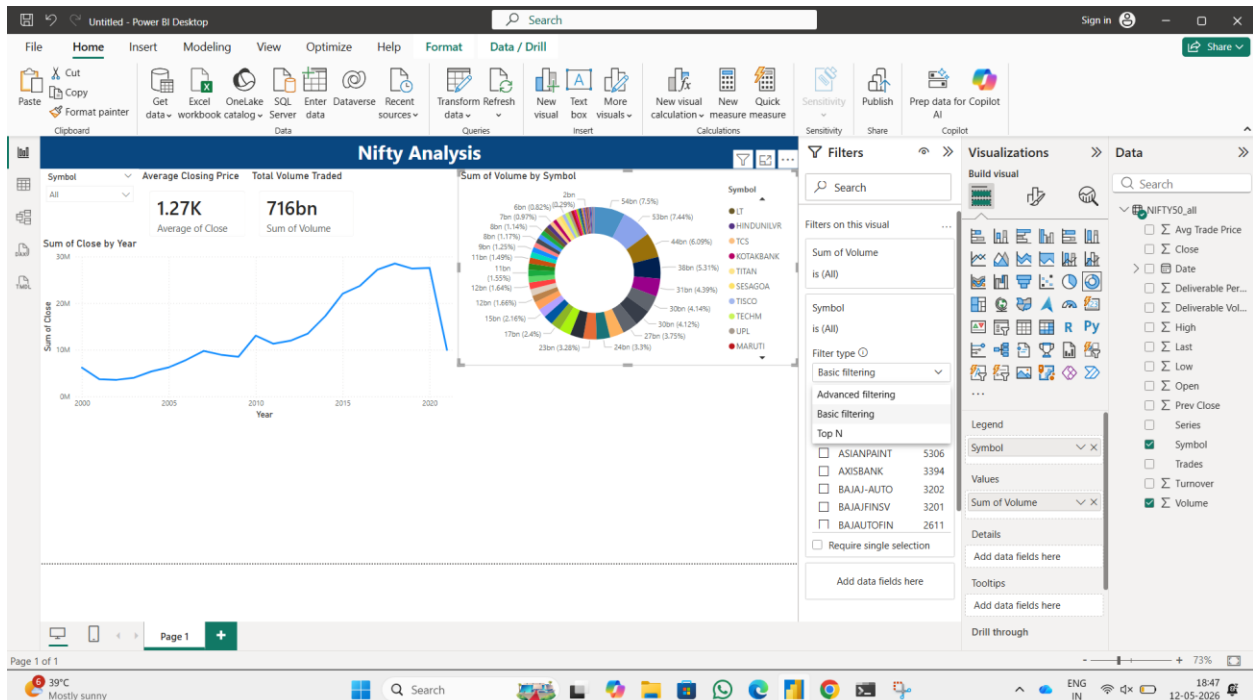
Look to the immediate left of the Visualizations pane. You will see a pane called **Filters**. (If it's collapsed, click the little arrow > to expand it)



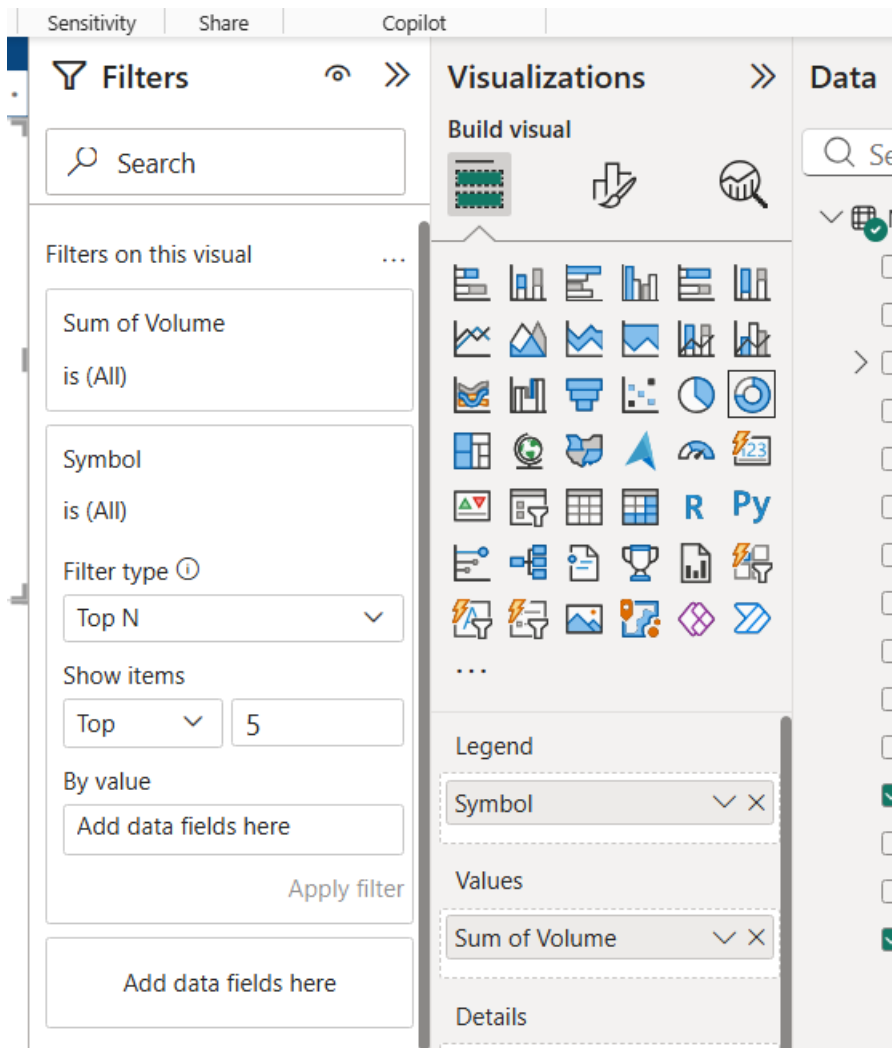
In the Filters pane, look under the section titled **"Filters on this visual"**. You will see your **Symbol** column listed there. Click the little down arrow v right next to the word "Symbol" to open its menu.



You will see a dropdown box that says **"Filter type"** (it is usually set to "Basic filtering"). Click that dropdown and change it to **Top N**.

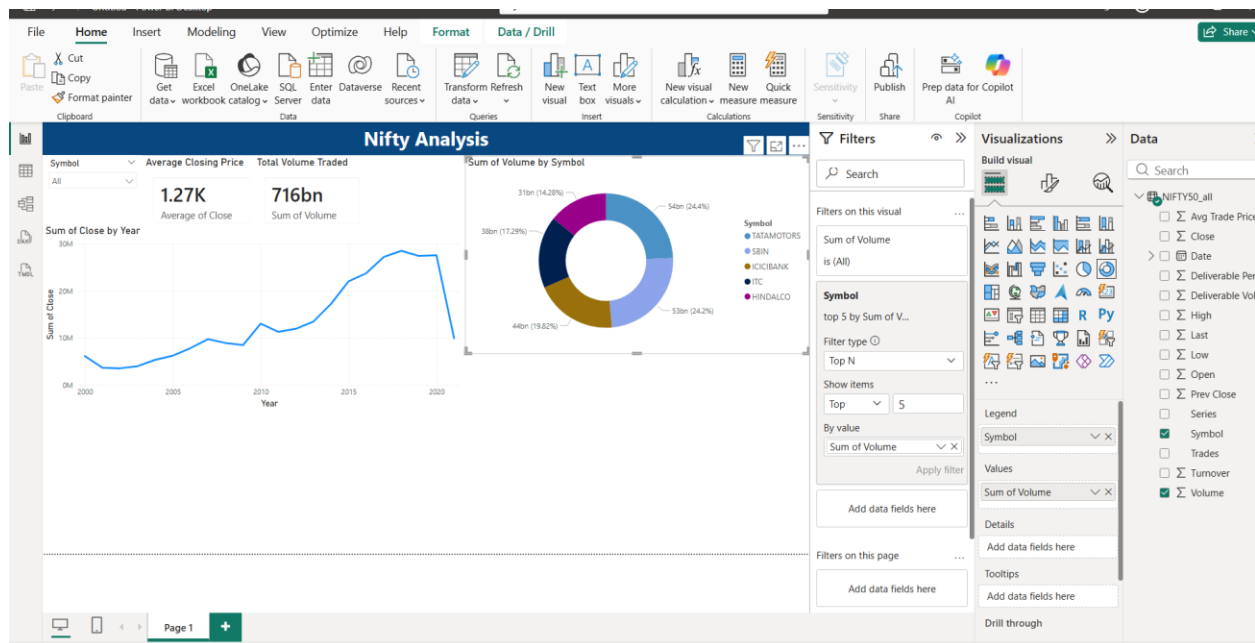


Right below that, under "Show items", you will see a box. Type the number **5** into that box.



Now, go all the way back to your **Data pane** on the far right. Click and drag your **Volume** column over to the Filters pane and drop it into the empty box that says "**By value**".

- ☐ Σ Avg Trade Price
- ☐ Σ Close
- ☐  Date
- ☐ Σ Deliverable Per.
- ☐ Σ Deliverable Vol.
- ☐ Σ High
- ☐ Σ Last
- ☐ Σ Low
- ☐ Σ Open
- ☐ Σ Prev Close
- ☐ Series
- ☒ Symbol
- ☐ Trades
- ☐ Σ Turnover
- ☒ Σ Volume



Step 16:-Click anywhere on the **blank white space** of your dashboard canvas (make sure no other charts are highlighted).

Look at the **Visualizations pane** on the right side.

Click the **Line chart** icon (it looks like a jagged line going up).

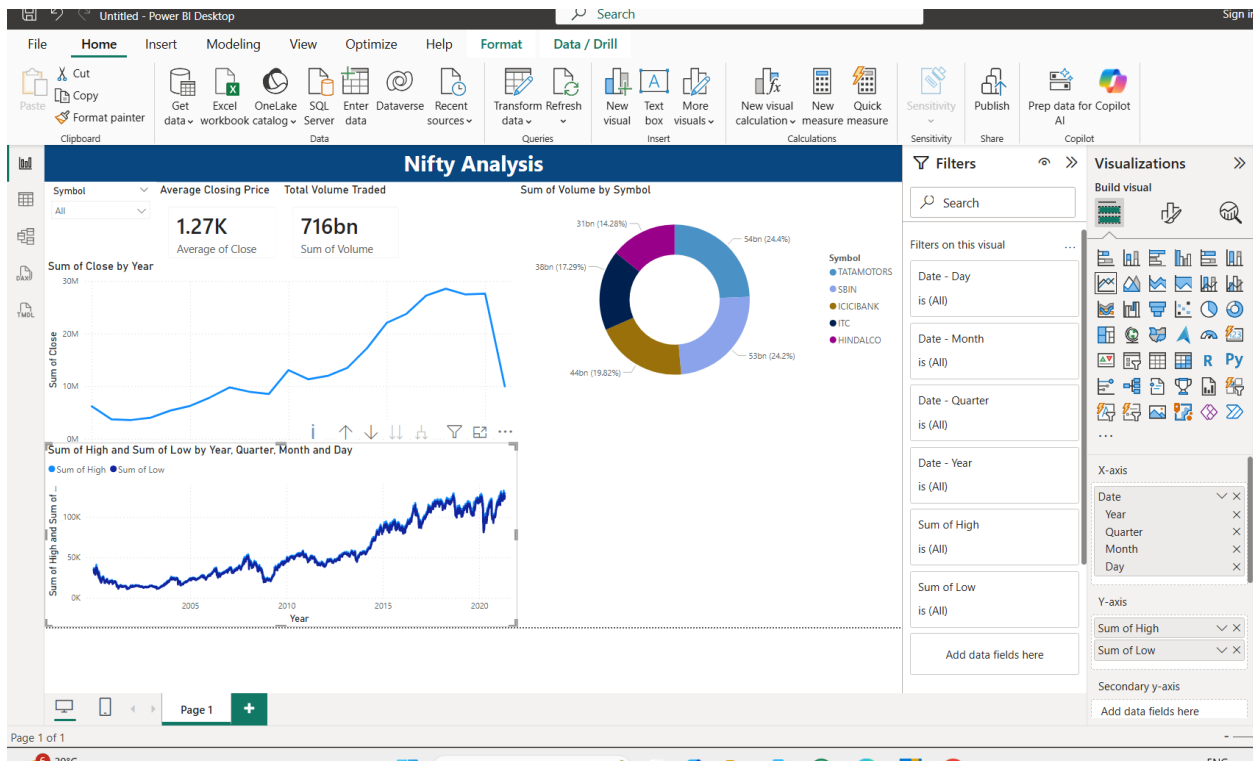
Resize the empty chart placeholder and place it wherever you have space on your dashboard.

Go to your **Data pane** on the far right.

Drag your **Date** column and drop it into the **X-axis** box in the Visualizations pane.

From the Data pane, drag the **High** column and drop it into the **Y-axis** box. (This is your Max line).

Now, from the Data pane, drag the **Low** column and drop it directly **underneath** the High column inside that exact same **Y-axis** box. (This is your Min line).



Step 17:-

Step 18:- Click on blank Canvas and Look at the **Visualizations** pane on the right side. Click the visual that has the numbers "123" on it (this is the **Card** visual). An empty square box will appear.

Go to your Data pane on the far right. Drag the **High** column and drop it into the **Fields** box in the Visualizations pane.

Right now, the card is probably adding up 14 years of high prices (showing a massive, useless number).

Click the little downward arrow (v) right next to the word "High" in the Add your fields here box.

Change it from **Sum** to **Maximum**. This will give you Max High

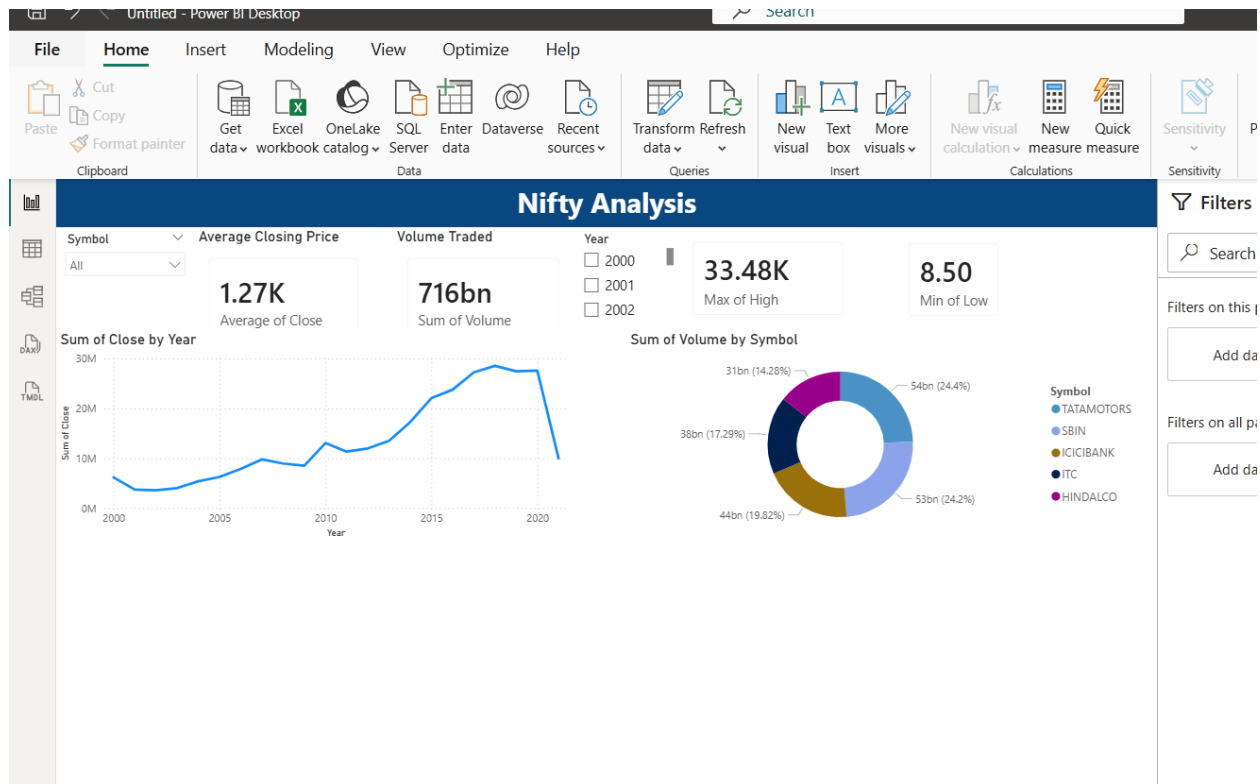
Now Click on blank Canvas. Click the visual that has the numbers "123" on it (this is the **Card** visual). An empty square box will appear.

Go to your Data pane on the far right. Drag the **Low** column and drop it into the **Fields** box in the Visualizations pane.

Click the little downward arrow (v) right next to the word "Low" in the Add your fields here box.

Change it from **Sum** to **Minimum**. This will give you Min Low

Step 19:- Click on Card Visual from Visualization Pane. Then drag Year from Data Pane into your Add your data fields then go to format your visual in Visualization Pane go to Slicer Settings and Style field select Vertical List

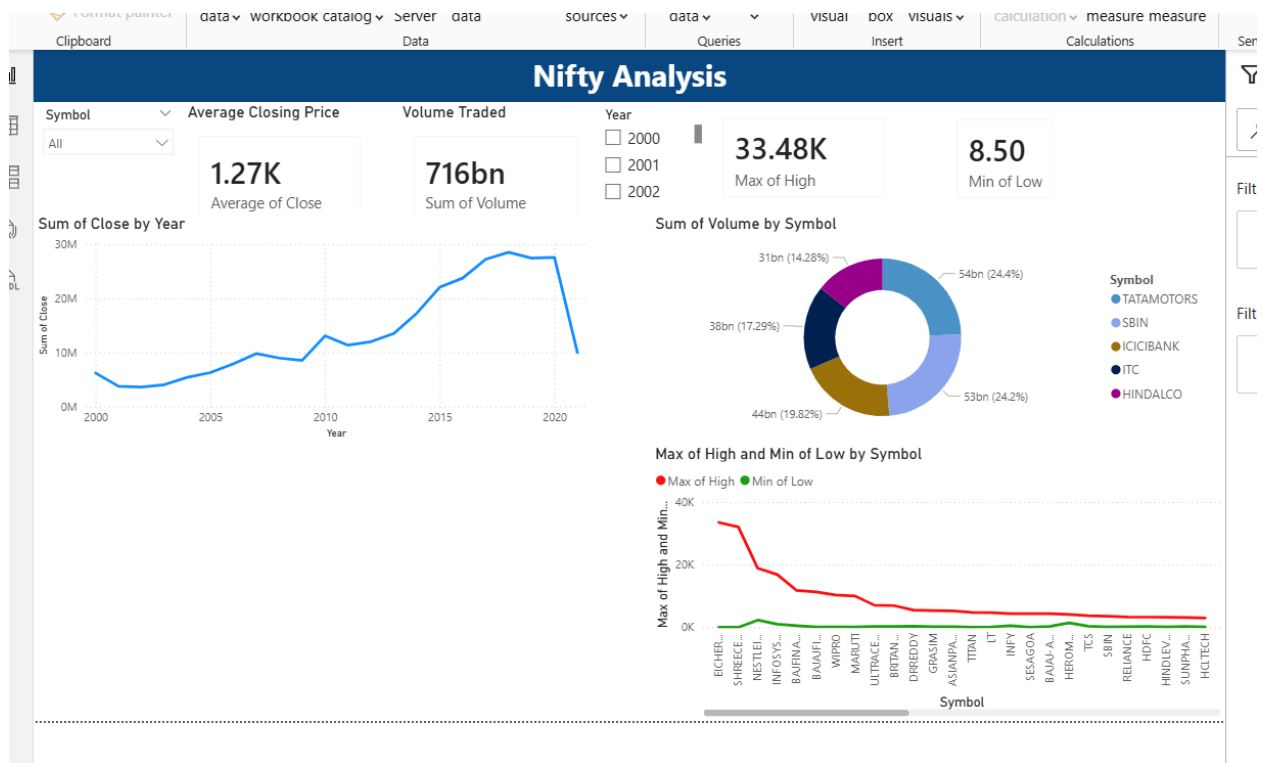


Step 20:- Click anywhere on the **blank white space** of your dashboard canvas (make sure no other charts are highlighted). Look at the **Visualizations pane** on the right side. Click on Line chart in Visualization Pane

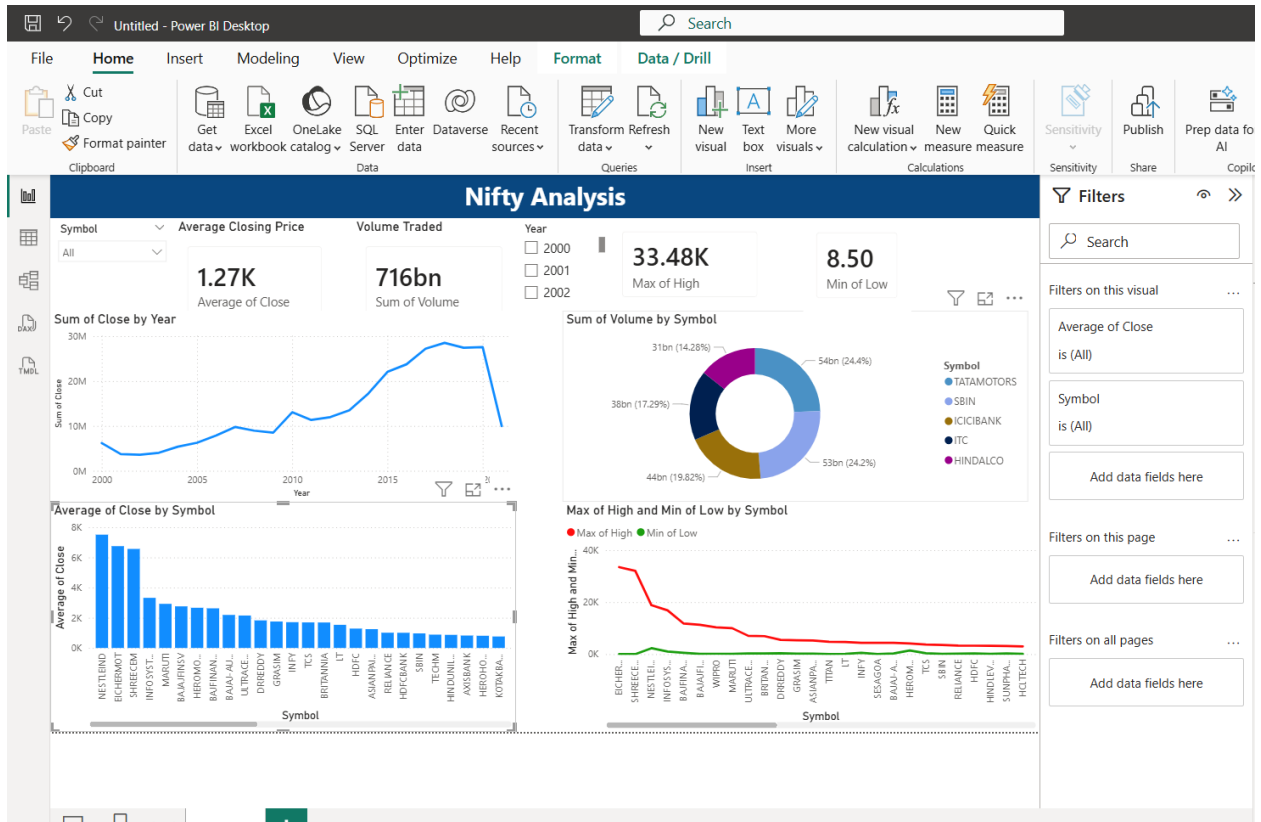
From Data panel: Symbol into: X-axis and Drag High into Y Axis. Click dropdown beside High and select maximum

Now drag Low from Data Pane to Y Axis. Ensure that Low is below High in Y Axis field and not above it. Click dropdown beside Low and select minimum

Go to **Format Visual** (paint brush icon) and click on Lines and expand it then click Series field and select Max of High and then change color to Red and then in Series field select Min of Low and change color to Green



Step 21:-Click on blank canvas white background.Click Clustered column chart from Visualization Pane.Now drag Symbol from Data Pane to X axis Field and Close to Y Axis. Click dropdown beside Close and select Average



Your final output will look like below:-

